
■■ RaniyaMart E-Commerce Platform

Full-Stack Java Web Application Presentation Deck

Project Identity: RaniyaMart (3-Tier MVC Web Application)

Tech Stack: Java 17, Servlets, JSP, JSTL, H2 DB, HikariCP, Maven, Render Cloud

Deployment URL: <https://raniyamart.onrender.com>

Repository: <https://github.com/Raniya08/RaniyaMart.git>

Currency Standard: Indian Rupees (₹)

1. Project Goal & Core Requirements

Core Objectives	Implemented Solution
• Multi-role authentication & authorization. • Secure online shopping in Indian Rupees (₹). • Persistent product inventory & order processing. • Zero broken image links across categories.	• Role-Based Access Control (Buyer, Seller, Admin). • Multi-channel Payment Checkout (UPI, Card, COD). • Automated category Unsplash image fallbacks. • 3-Tier MVC Architecture with H2 Relational DB.

2. System Architecture (3-Tier MVC)

Layer	Technologies Used	Responsibilities
Presentation (View)	JSP, JSTL, HTML5, CSS3, JavaScript	Renders responsive product grids, checkout forms, and user dashboards.
Controller Layer	Java Servlets (`HttpServlet`)	Handles HTTP requests, session management, routing, and form validation.
Service Layer	Java Interfaces & Impl classes	Executes business logic (`ProductService`, `OrderService`, `UserService`).
Data Layer (DAO)	JDBC, HikariCP, H2 SQL Database	Manages DB connection pool, executes parameterized SQL queries.

3. Role-Based Access Control (RBAC)

Role	Default Credentials	Key Privileges & Capabilities
ADMIN	`admin@raniyamart.com` `Password123!`	• Global user management & role breakdown. • System metrics oversight & catalog monitoring. • Protected single admin seed account.
SELLER	`seller@raniyamart.com` `Password123!`	• Product listing creation, editing & deletion. • 1-click sample image selection. • Incoming buyer order status workflow updates.
BUYER	`buyer@raniyamart.com` `Password123!`	• Catalog search & category filtering. • Shopping cart management. • Checkout via UPI, Card, or Cash on Delivery.

4. Checkout & Payment Module Engine

Payment Mode	Details & Validation	User Experience Flow
■ UPI Payment	GPay, PhonePe, Paytm, BHIM UPI. Validates Virtual Payment Address (VPA).	Dynamic radio selection, displays app choice & UPI ID on order confirmation.
■ Credit / Debit Card	Visa, MasterCard, RuPay. Validates cardholder name, expiry, CVV.	Extracts last 4 digits (e.g. `Ending in *8892`) for privacy & receipt formatting.
■ Cash on Delivery (COD)	Doorstep cash or UPI QR scan upon delivery.	Instant order confirmation with zero gateway delay.

5. Smart Product Image & Fallback Engine

Feature	Implementation & Logic
Category Auto-Fallback	When `imageUrl` is blank, `ProductServiceImpl.java` inspects category and product title keywords (e.g., *Sunscreen, Electronics, Mobile, Fashion, Furniture, Books*) to auto-assign high-res Unsplash images.
URL Normalization	`SellerServlet.java` inspects user-pasted image links. If `https://` is missing (e.g. `www.example.com/...`), it automatically normalizes the link to full `https://` format.
1-Click Sample Pickers	`seller/dashboard.jsp` & `product-form.jsp` include 1-click sample buttons (■ Sunscreen, ■ Phone, ■ Laptop, ■ Watch, ■ Furniture).

6. Database Design & Security

Database Feature	Technical Details
Database Engine	H2 Relational Database running in file-persisted mode (<code>./data/raniyamart.mv.db</code>) with MySQL compatibility.
Connection Pooling	HikariCP (<code>HikariDataSource</code>) maintains a high-performance pool of reusable JDBC connections.
SQL Security	100% parameterization using <code>PreparedStatement</code> to eliminate SQL injection vulnerabilities.
Password Hashing	BCrypt password hashing algorithm (<code>jbcrypt</code>) for safe credential storage.
Cloud Env Support	<code>DBUtil.java</code> checks <code>DB_URL</code> , <code>DB_USER</code> , <code>DB_PASSWORD</code> env vars for connecting to cloud databases (PostgreSQL/MySQL).

7. Terminal Demonstration Commands (Viva Guide)

Target Query	PowerShell / Terminal Execution Command
Display All Users	java -cp "C:\Users\Hi\.m2\repository\com\h2database\h2\2.2.224\h2-2.2.224.jar" org.h2.tools.Shell -url "jdbc:h2:./data/raniyamart" -user sa -sql "SELECT id, full_name, email, role FROM users;"
Display Products List	java -cp "C:\Users\Hi\.m2\repository\com\h2database\h2\2.2.224\h2-2.2.224.jar" org.h2.tools.Shell -url "jdbc:h2:./data/raniyamart" -user sa -sql "SELECT id, name, category, price, stock_qty FROM products;"
Launch Web GUI Console	java -cp "C:\Users\Hi\.m2\repository\com\h2database\h2\2.2.224\h2-2.2.224.jar" org.h2.tools.Console

8. Quality Assurance & Cloud Deployment

Phase	Details & Metrics
Automated Unit Testing	JUnit 5 + Mockito unit test suite verifying `UserService`, `ProductService`, `CartService`, `OrderService`, `NotificationService`, `ChatService`, and `UserDAO`. Result: 19 / 19 Tests Passed (0 Failures, 0 Errors).
Build Tooling	Apache Maven (`mvn clean test package`) compiling 60 Java source files into standalone WAR deployment artifact.
Cloud Host	Render.com Cloud Web Service with Git-triggered auto-deployment. Live App URL: https://raniyamart.onrender.com

9. Conclusion & Project Links

Key Project Deliverables Achieved:

- ✓ Complete 3-Tier Java MVC E-Commerce Architecture.
- ✓ Role-Based Access Control (Admin, Seller, Buyer).
- ✓ Multi-channel Payment Gateway Engine (UPI, Card, COD).
- ✓ Category Image Auto-Fallback & Sunscreen Recovery.
- ✓ 100% Automated Unit Test Suite (19/19 Passed).
- ✓ Live Render Cloud Deployment.

Resource	Link / Access Information
Deployed Web Application	https://raniyamart.onrender.com
GitHub Source Repository	https://github.com/Raniya08/RaniyaMart.git
Default Demo Credentials	Admin: `admin@raniyamart.com` Seller: `seller@raniyamart.com` Buyer: `buyer@raniyamart.com` Password for all accounts: `Password123!`